Birds of a Feather Don't Flock Together:

China's Foreign Aid Targeting and Governance Style

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Independent Analysis Project

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GPCO 454 - Quantitative Methods II

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Memo

Date: March 11th 2026

To: Marco Rubio, Secretary of State

From: Edgar Agunias

Regarding: China's Aid Targeting Strategy

Introduction

With a large US pull back on foreign aid contributions in the UN and otherwise, many in the international community look to China to fill this gap. Many see it logical for China to fill this gap in influence, however traditionally China has been selective in where it allocates its ODA. This research was designed to answer if China's foreign aid is tightly correlated to more authoritarian governments. China's aid allocation differs from the US which is typically concentrated in the UN, while China's is more direct and thus bilateral. Chinas development finance and belt and road initiative is seen more politically motivated in the West, when compared to Western aid. Therefor the purpose of this research is to see if China's Aid disbursement method has preference to other authoritarian governments. After extensive quantitative research using a country level cross sectional regression analysis, there is a positive association. However, after economic and demographic controls are accounted for in the full sample, the countries government does not have a statistically significant relationship with how much aid a country receives. At the same time, among countries that did receive positive Chinese aid, a one point increase in autocracy score is associated with about 2.95 times higher mean annual aid. Suggesting China may have different criterion for aid disbursement if aid is being given.

Theory and Hypothesis:

Our causal research question is: Do more autocratic regimes receive more funding from China on average?

Null Hypothesis: Autocratic Regimes do not differ in average aid received when compared to Democratic ones.

Alternative Hypothesis: Autocratic regimes receive more aid on average when compared to Democratic ones.

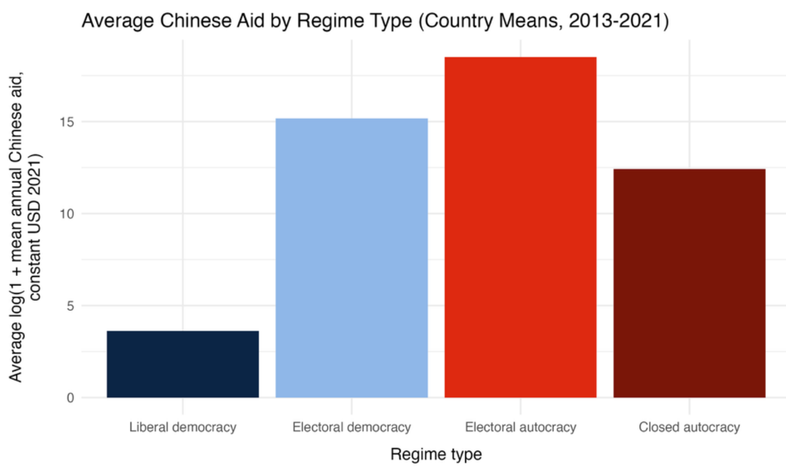
$$H_0\colon \mu_{Autocratic} = \mu_{Democratic} \qquad H_a\colon \mu_{Autocratic} > \mu_{Democratic}$$

Model Construction:

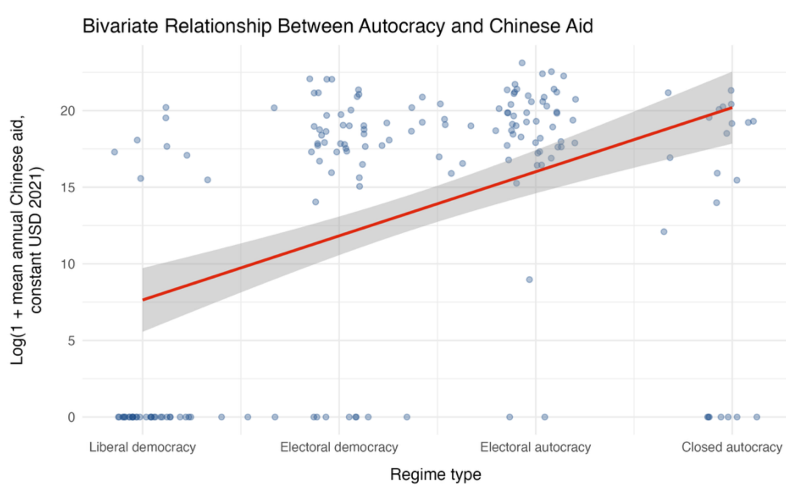
$$\log\left(1 + \overline{\text{Chinese Aid}}_c\right) = \beta_0 + \beta_{1\overline{\text{Autocracy}}_c} + \beta_2 \log\left(\overline{\text{GDP per capita}}_c\right) + \beta_{3\overline{\text{CPI}}_c} + \beta_{4\overline{\text{Extreme Poverty}}_c} + \beta_{5\overline{\text{Gini}}_c} + \varepsilon_c$$

Data Description:

Using the Our World in Data (OWID) data set for classification of autocratic governments and the Chinese Global Development Finance Data set for levels of foreign Aid provided by China to run my cross sectional regression. Our final data set collapsed the years of 2013 to 2021 into country averages, making the unit of analysis the country. The bivariate model utilized 174 countries, the controls model utilized 135 countries, and the positive aid only model utilized 92 countries. Our reconfigured OWID data set classified autocratic governments on a scale from 0 to 3, with 0 being electoral democracies and 3 being closed autocracies. China itself is classified as a closed autocracy. In the graph below we can see that even when adjusting aid as a logarithm of its mean annual value so that extremely large dollar donations do not dominate the graph, more autocratic regimes received more aid from China. We also see that after a regression analysis there is a positive relationship between autocracy levels and level of aid provided in the bivariate model, though in the full controls model this relationship loses statistical significance.



Descriptive figure. Each bar reports the mean of log(1 + mean annual Chinese aid in constant 2021 USD) across country-level observations created by averaging country-year values from 2013 to 2021.



Relational figure. Points are countries collapsed to one observation each using 2013-2021 means. The line is an OLS fit with 95% confidence interval.

Chinese Aid and Regime Type (Country-Level Cross-Section)

	Dependent variable:		
	Log(1 + mean annual Chinese aid in constant USD 2021)		
	Bivariate	Controls	Controls, positive-aid only
	(1)	(2)	(3)
Autocracy score	4.186***	0.948	1.083***
	(0.769)	(1.171)	(0.397)
Log GDP per capita		-2.591***	0.812***
		(0.960)	(0.276)
Corruption Perceptions Index		-0.207***	-0.009
		(0.061)	(0.026)
Extreme poverty share		-0.111**	0.022
		(0.046)	(0.013)
Gini index		29.888***	1.008
		(8.620)	(3.314)
Constant	7.642***	35.808***	9.717***
	(1.165)	(9.093)	(2.533)
Years collapsed	2013-2021 mean	2013-2021 mean	2013-2021 mean
Sample	All countries	All countries	Aid recipients only
Standard errors	HC1	HC1	HC1
Observations	174	135	92
R2	0.202	0.594	0.165
Adjusted R2	0.197	0.578	0.117
Note:			*p<0.1; **p<0.05; ***p<0.01

Methodology and Findings:

Our regression structure is a simple cross sectional linear regression. After controlling for variables this positive relationship loses statistical significance in the full sample. For example, GDP per capita to control for development levels of economies and their correlated effects on governmental structures. Corruption index was used to control for government quality, as perhaps China utilizes it to assess efficacy of their aid which may be lost to corruption. Extreme poverty share acts as a control for aid need in a country. As countries with higher percentage of the population living in extreme poverty are in greater need of aid. Gini coefficient controlled for level of inequality within a country receiving aid. Finally, because the data was collapsed into country averages from 2013 to 2021, the model compares countries on their mean aid levels rather than controlling for country specific and year specific changes in aid disbursement. All this to isolate any correlational relationship between style of government and the corresponding level of Chinese aid. We chose to make a distinction between countries who received Chinese aid and those that did not to prevent selection bias within our sample.

After all controls were taken into account in the full sample, it is clear that there is no statistically relevant correlational relationship between level of autocracy and amount of Chinese aid. However, among countries that did receive aid, the autocracy score remains statistically significant, and a one point increase in autocracy score is associated with about 2.95 times higher mean annual aid. The correlation coefficient of Autocracy score is statistically significant at the 99% confidence interval. And when comparing model 3 to model 1 the r squared These findings show that within the full sample China does not appear to allocate aid differently by regime type, but among recipient countries how the autocratic level of the government is a significant factor in the amount of aid provided.

Limitations and Conclusion:

Some limitations of research were that GDP of countries could not be taken into account as only GDP per Capita was provided within the data set. This could introduce a bias for larger populations or economies which China may be more inclined to provide aid or shy away from providing aid. Another limitation is that by collapsing the data into a country level cross section, year to year variation is no longer captured. The full controls model also loses observations because of missing control variables, and some country cases appear to have stronger influence on the estimates than others. Therefore, while the full sample does not show a statistically significant relationship between autocracy and Chinese aid, the positive aid model indicates that if a country is more autocratic and has received aid at all that on average their mean aid increases by 2.95 times all things remaining constant. Lastly, the R squared value, or the explanatory power of the 3rd model accounts for 16.5% of the variance in the amount of aid disbursed by the Chinese. However, this only applies to countries which have received aid in the past. There is evidence to reject the null hypothesis that democratic countries received equal aid to autocratic ones. Areas for further research could be the industrial make up of countries receiving aid, and whether more wealthy countries receive aid. Chinas aid disbursement strategy seems at least in some part related to how similar their government styles are constructed. As a counter factual, the United States could also be investigated to determine whether this is out of the norm for nations which provide aid.

Appendix

Gauss Markov Tests

VIF Test
VIF was below 5 in all variables. Little evidence of Multicolinearity

autocracy_score	2.01801820934758
log_gdp_pc	4.4901424736068
cpi	3.61815701518997
extreme_poverty	2.60982785962193
gini	1.25280965931702

Breush-Pagan Test
We controlled for this by utilizing robust standard errors, however there was no sign of heteroskedasticity

model	statistic	df	p_value
CS_M1_bivariate	0.994273516314578	1	0.318700128093471
CS_M2_controls	5.78367876657589	5	0.327840230150812
CS_M2_controls_positive_aid	11.0460322403648	5	0.0504748999685814

Outliers
These variables were subsequently dropped if they exceeded the Cooks Distance lines. Or were overleveraged and distance in fits.

Model observations 135

k (predictors) 5

Threshold |studentized residual| > 2

Threshold leverage > 0.088889

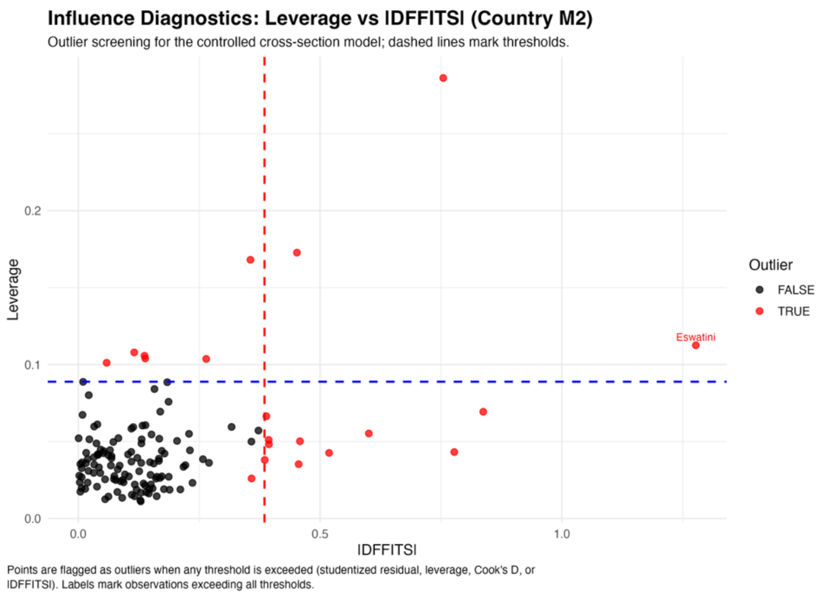
Threshold Cook's D > 0.031008

Threshold |DFFITS| > 0.384900

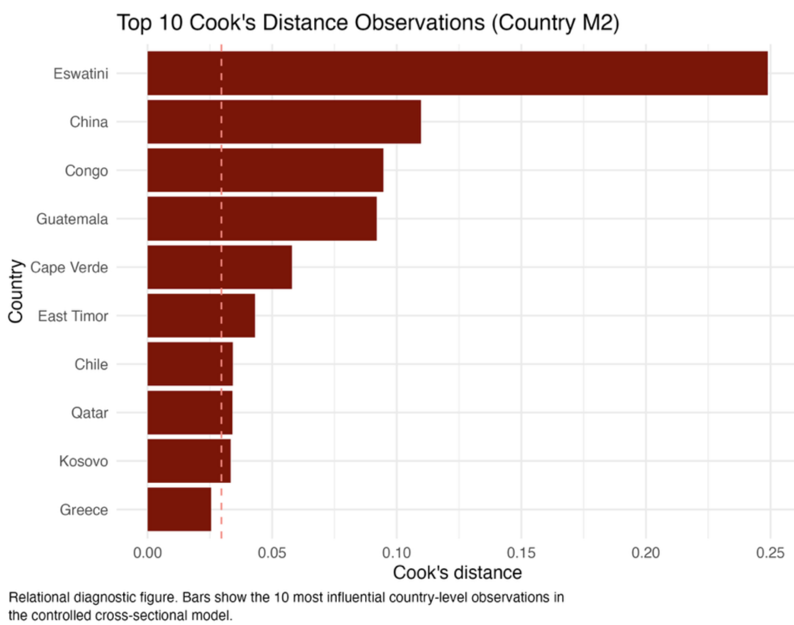
Flagged outliers (any threshold) 20

Flagged egregious outliers (all thresholds) 1

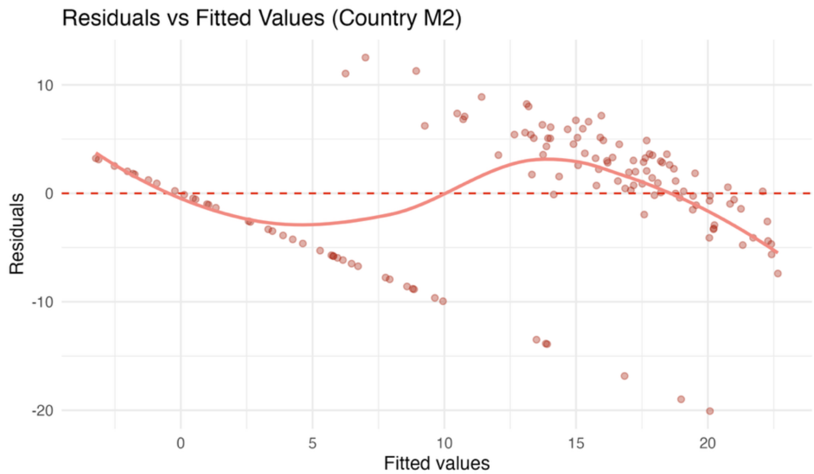
Graphs:



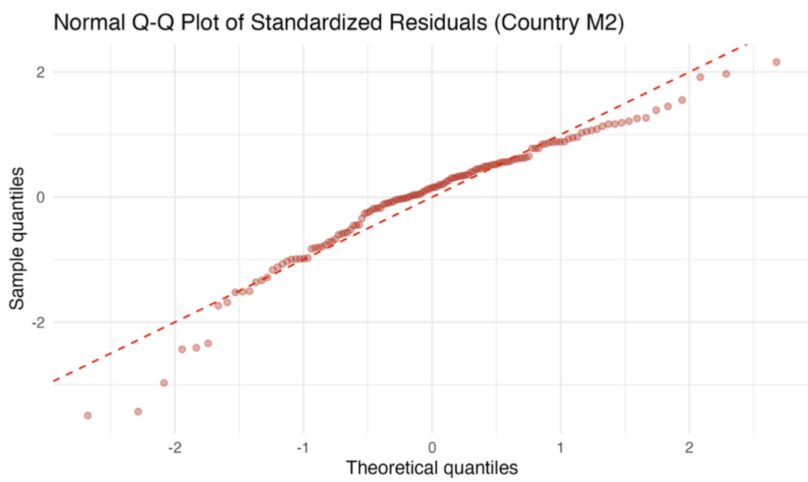
Influence Diagnostics: Leverage vs |DFFITS| (Country M2)
Outlier screening for the controlled cross-section model; dashed lines mark thresholds.
Leverage
0.0
0.1
0.2
0.0
0.5
1.0
|DFFITS|
Eswatini
Outlier
FALSE
TRUE
Points are flagged as outliers when any threshold is exceeded (studentized residual, leverage, Cook's D, or |DFFITS|). Labels mark observations exceeding all thresholds.



Top 10 Cook's Distance Observations (Country M2)
Country
Eswatini
China
Congo
Guatemala
Cape Verde
East Timor
Chile
Qatar
Kosovo
Greece
0.00
0.05
0.10
0.15
0.20
0.25
Cook's distance
Relational diagnostic figure. Bars show the 10 most influential country-level observations in the controlled cross-sectional model.



Relational diagnostic figure. Residuals from the controlled country-level model are plotted against fitted values to assess functional form and heteroskedasticity.



Relational diagnostic figure. Standardized residuals from the country-level controlled model are compared with normal quantiles.